

ABDUL WALI KHAN UNIVERSITY MARDAN



CURRICULUM

of { BS Computer Science
Associate Degree Computing }
(2026)



Department of Computer Science

TABLE OF CONTENTS

1. INTRODUCTION.....	3
1.1. VISION AND MISSION	3
1.2. ACADEMIC RESOURCES	4
1.3. FACULTY	4
2. UNDERGRADUATE COURSES.....	4
2.1. COURSE CATEGORIZATION AND DISTRIBUTION.....	4
2.2. DEFINITION OF COURSE CODES	5
2.3. GENERAL EDUCATION COURSES	5
2.4. INTERDISCIPLINARY/ALLIED COURSES	6
2.5. FUNDAMENTAL COMPUTING OR MATHEMATICAL COURSES	7
2.6. COMPUTER SCIENCE CORE COURSES	7
2.7. MANDATORY LAB COURSES	7
2.8. PROFESSIONAL COURSES	8
2.9. SPECIALIZATION ELECTIVE COURSES (GENERAL COMPUTER SCIENCE OR HONORS)	8
2.10. SPECIALIZATION ELECTIVE COURSES (SOFTWARE)	8
2.11. SPECIALIZATION ELECTIVE COURSES (ARTIFICIAL INTELLIGENCE).....	8
2.12. SPECIALIZATION ELECTIVE COURSES (CYBER SECURITY)	9
2.13. ALL SPECIALIZATION COURSES IN A SEQUENCE	9
2.14. FYP EVALUATION CRITERIA	9
2.15. INTERNSHIP EVALUATION CRITERIA	10
3. ASSOCIATE DEGREE - COMPUTING	10
3.1. GENERAL EDUCATION COURSES	10
3.2. FUNDAMENTAL COMPUTING OR MATHEMATICAL COURSES	11
3.3. COMPUTING CORE COURSES.....	11
3.4. MANDATORY LAB COURSES	12
3.5. POOL OF ELECTIVE COURSES	12
4. PROGRAM'S DETAILS.....	12
4.1. PROGRAM LEARNING OUTCOMES FOR BS COMPUTER SCIENCE	12
4.2. STANDARD NOMENCLATURE	13
4.3. ELIGIBILITY CRITERIA.....	13
4.4. ASSOCIATE DEGREE PATHWAY (ENTRY, EXIT, AND TRANSFER REQUIREMENTS).....	14
4.5. IMPLEMENTATION.....	15
4.6. COURSE DESCRIPTION.....	15

5.	SEMESTER WISE COURSE DISTRIBUTION.....	16
5.1.	SEMESTER WISE COURSE DISTRIBUTION (BS COMPUTER SCIENCE – ALL DICIPLINES)	16
5.2.	SEMESTER WISE COURSE DISTRIBUTION (BS COMPUTER SCIENCE - HONORS)	17
5.3.	SEMESTER WISE COURSE DISTRIBUTION (BSCS – SOFTWARE)	18
5.4.	SEMESTER WISE COURSE DISTRIBUTION (BSCS – ARTIFICIAL INTELLIGENCE)	19
5.5.	SEMESTER WISE COURSE DISTRIBUTION (BSCS – CYBER SECURITY)	20
5.6.	SEMESTER WISE COURSE DISTRIBUTION (ASSOCIATE DEGREE)	21

1. INTRODUCTION

Recent developments in computer hardware, software and communication technologies have offered new exciting opportunities and challenges for the creation of innovative learning environments for Computer Science and its curricula design. One of the key elements here is to prepare the graduates for the future. The challenge of getting all newly emerging technologies incorporated into the curriculum is becoming pivotal for the effectiveness of curricula. There is a need for curricula structures that can grow as we put new demands on them. The curriculum is required to provide integration of all components and the foundations that allow accessing all the new knowledge and technology to fulfil the vision of the future.

The basic intention of an academic programme in Computer Science is to develop the students' critical, professional thinking and intuition. The curriculum must be structured to provide a balanced mixture of learning experiences, to make our graduates capable and sound professionals. As a result, a graduate should be able to assume responsible positions in business, government, research, development, and planning. The programme should also provide an excellent foundation for further learning and training. The Computer Science curriculum is expected to provide such an environment to put into practice the principles and techniques learnt during the course work academic programme.

1.1. VISION AND MISSION

a. Vision Statement

To be one of the Top Universities of the country striving to have an impact on the global Scene.

b. Mission Statement

To empower a diverse community of learners by involving all stakeholders thus providing them with opportunities to excel in their lives and professional careers by creating an enabling learning environment focusing on quality education, fostering impactful research, promoting innovation for growth and imparting skills through experience and collaboration.

1.2. ACADEMIC RESOURCES

Resources	Objectives	Major Equipment
3 Computing Labs	These computing labs are used for lab work of various courses.	Computer systems with network connectivity
Classrooms	20 classrooms used by undergraduate and graduate students	
Theater Hall	3 Theater halls, used for seminars, presentations and other similar activities	Multimedia projector and sound system
Other facilities	Girls common room, staff meeting room, washrooms, stores	None
Incubation Center	Used for student startups and industrial linkages	Computer systems with network connectivity

1.3. FACULTY

S#	Name	Designation
1.	Prof. Dr. Sher Afzal Khan	Dean P&NS/Professor
2.	Prof. Dr. Nadeem Iqbal	Chairman/ Professor
3.	Prof. Dr. Maqsood Hayat	Professor
4.	Prof. Dr. Rahim Khan	Professor
5.	Dr. M. Irfan Uddin	Associate Professor
6.	Dr. Muhammad Tahir	Assistant Professor
7.	Dr. Hashim Ali	Assistant Professor
8.	Dr. Aftab Ahmed	Lecturer
9.	Dr. Ashraf Zia	Lecturer
10.	Dr. Nazia Azim	Lecturer
11.	Dr. Muhammad Shahzad	Lecturer
12.	Mr. Shahzad Alam	Demonstrator

2. UNDERGRADUATE COURSES

2.1. COURSE CATEGORIZATION AND DISTRIBUTION

Areas	HEC Proposed		AWKUM DCS BS Program		AWKUM DCS BS Program (Associate)	
	CHrs	Courses	CHrs	Courses	CHrs	Courses
General Education	34	14	34	15*	34	15*
Computer Sc. Core	48	14	48	14	26	7
Specialization Electives	24	8	27	9	13	4
Professional Certification	3	1	3	1	--	--
Inter./Allied Courses	12	4	12	4	--	--
Internship	3	1	3	1	--	--
Capstone Project / FYP	6	1	6	2*	--	--
Totals	130	43	133	46	73	26

*Fahm-e-Quran and Final Year Project (FYP) are each offered in two parts (I & II). One additional specialization course is included to further enhance students' skills.

2.2. DEFINITION OF COURSE CODES

Course Category	Course Code Pattern
General Education Courses	Parent departments
Interdisciplinary/Allied Courses	Parent departments
Computer Science Core Courses	CS###
Specialization Electives	
1. General Computer Science (Honors)	CS###
2. Software	CS###
3. Artificial Intelligence	CS###
4. Cyber Security	CS###

Each course code consists of a two-letter prefix followed by a three-digit number (###). The coding scheme is applied to all the courses offered at the department, including basic, core, specialization electives, and certifications. These courses are identified with the prefix CS.

The three digits in the course code represent the following:

- The first digit indicates the academic level/year of the course (1–4).
- The second digit indicates the category of the subject.
- The third digit represents the sequence number of the course within the same category and academic year.

Courses in General Education and the Interdisciplinary /Allied courses are assigned course codes by their respective departments.

The categorization of subjects based on the second digit of the course code is defined in the table below.

Digit	Category Type	Description
0	Basic Computer or Mathematics	Basic requirements in computing or mathematics.
1	Computer Science Core	All core courses of the computer science discipline.
2	Comp Science Specialization	Specializations courses of general computer science or (honors)
3	Software Specialization	Specialization courses of software discipline.
4	Artificial Intelligence Specialization	Specialization courses of artificial intelligence discipline.
5	Cyber Security Specialization	Specialization courses of cyber security discipline.
6	Associate Degree Computing	Only associate degree specific courses.
9	Projects	FYPs, Professional Courses, and Internships.

2.3. GENERAL EDUCATION COURSES

S#	Code	Course Title	Sub-Category	Credit Hrs	Pre-Req
1	XX###	Applied Physics*	Natural Sciences	3(2+1)	

		<i>(Choose only ONE Course)</i> -Introductory Mechanics -Modern Physics -Introductory Electricity -Everyday Science			
2	XX###	Functional English	ENG	3(3+0)	
3	XX###	Expository Writing	ENG	3(3+0)	
4	XX###	Applied Mathematics for Computing I*	QR-I	3(3+0)	
5	XX###	Applied Mathematics for Computing II*	QR-II	3(3+0)	
6	XX###	Application of ICT	-	3(2+1)	
7	XX###	Introduction to Management*	Social Sciences	2(2+0)	
		<i>(Choose any ONE Course)</i> -Economics			
8	XX###	Professional Practices*	Arts & Humanities	2(2+0)	
		<i>(Choose any ONE Course)</i> <i>-(Any other Arts & Humanities Course)</i> Introduction to Journalism Introduction to Philosophy			
9	XX###	Entrepreneurship	-	2(2+0)	
10	XX###	Civics and Community Management	-	2(2+0)	
11	XX###	Pakistan Studies	-	2(2+0)	
12	XX###	Islamic Studies / Ethics	-	2(2+0)	
13	XX###	Ideology and Constitution of Pakistan	-	2(2+0)	
14	XX###	Fehm e Quran I	-	1(1+0)	
15	XX###	Fehm e Quran II	-	1(1+0)	

*As per HEC guidelines, there should be a list of options to select courses, but for the smooth administration of teaching and examination (including affiliated colleges), we are restricting that to only offering these courses.

2.4. INTERDISCIPLINARY/ALLIED COURSES

S#	Code	Course Title	Credit Hrs	Pre-Req.
1	XX###	Calculus and Analytical Geometry	3(3+0)	
2	XX###	Linear Algebra	3(3+0)	
		<i>Choose any TWO Courses</i>		
3	XX###	Discrete Structure*	3(3+0)	
4	XX###	Probability & Statistics*	3(3+0)	
		<i>(Choose any TWO Courses)</i>		
		Economics		
		Organizational Behavior		
		Media and Information Literacy		
		E-Governance and ICT Policy		
		Environmental Science		
		Professional Practice		
		Digital Marketing & E-Commerce		
		Technology Management		
		Financial Technology Fundamentals		
		Geographic Inf. Systems (GIS) & Spatial Data		
		Bioinformatics		

		Smart Cities and Urban Informatics Cyber Law & Digital Policy Technology and Sustainable Development Digital Media & Communication Studies Project Management Educational Technology		
--	--	---	--	--

*As per HEC guidelines, multiple course options are permissible; however, for smooth administration of teaching and examinations (including affiliated colleges), only the listed courses are offered.

2.5. FUNDAMENTAL COMPUTING OR MATHEMATICAL COURSES

S#	Code	Course Title	Cat. Type	Credit Hrs	Pre-Req.
1.	CS101	Application of ICT	0	3 (2+1)	
2.	CS102	Applied Mathematics for Computing I	0	3 (3+0)	
3.	CS103	Applied Mathematics for Computing II	0	3 (3+0)	CS102

2.6. COMPUTER SCIENCE CORE COURSES

S#	Code	Course Title	Cat. Type	Credit Hrs	Pre-Req
1.	CS111	Programming Fundamentals	1	4 (3+1)	
2.	CS112	Object Oriented Programming	1	4 (3+1)	CS111
3.	CS113	Digital Logic Design	1	4 (3+1)	
4.	CS211	Data Structures	1	4 (3+1)	CS111
5.	CS212	Database Systems	1	4 (3+1)	
6.	CS213	Software Engineering	1	3 (3+0)	
7.	CS214	Design and Analysis of Algorithms	1	3 (3+0)	
8.	CS215	Operating Systems	1	4 (3+1)	CS101
9.	CS216	Computer Networks	1	3 (3+0)	
10.	CS311	Artificial Intelligence	1	3 (3+0)	CS101
11.	CS312	Theory of Automata	1	3 (3+0)	CS103
12.	CS411	Information Security	1	3 (3+0)	CS103
13.	CS412	Computer Organization and Architecture	1	3 (2+1)	CS113
14.	CS413	Cloud Computing	1	3 (3+0)	CS101

2.7. MANDATORY LAB COURSES

S#	Code	Course Title	Cat. Type	Credit Hrs	Co Pre-Req.
1.	CS111L	Programming Fundamentals	1	1 (0+3)	CS111
2.	CS112L	Object Oriented Programming	1	1 (0+3)	CS112
3.	CS113L	Digital Logic Design	1	1 (0+3)	CS113
4.	CS211L	Data Structures	1	1 (0+3)	CS211
5.	CS212L	Database Systems	1	1 (0+3)	CS212
6.	CS215L	Operating Systems	1	1 (0+3)	CS215
7.	CS412L	Computer Organization and Architecture	1	1 (0+3)	CS412

2.8. PROFESSIONAL COURSES

S#	Code	Course Title	Cat. Type	Credit Hrs	Pre-Req.
1.	CS496	Professional Certification	9	3(0+3)	
2.	CS497	Internship	9	3(0+3)	
3.	CS498	Final Year Project - I	9	3(0+6)	
4.	CS499	Final Year Project - II	9	3(0+6)	

*The Final Year Project is offered in two parts, FYP-I and FYP-II, each evaluated separately with 100 marks.

2.9. SPECIALIZATION ELECTIVE COURSES (GENERAL COMPUTER SCIENCE OR HONORS)

Sp. #	Code	Course Title	Cat. Type	Credit Hrs	Pre-Req.
I.	CS321	Web Technologies	2	3(3+0)/3(2+1)	
II.	<<	Machine Learning	4	3(3+0)/3(2+1)	
III.	CS323	Advanced Programming	2	3(3+0)/3(2+1)	
IV.	CS324	Web Engineering	2	3(3+0)/3(2+1)	
V.	<<	Network Security	5	3(3+0)	
VI.	<<	Deep Learning	4	3(3+0)/3(2+1)	
VII.	CS423	Mobile Application Development	2	3(3+0)/3(2+1)	
VIII.	CS424	Internet of Things and Its Applications	2	3(3+0)/3(2+1)	
IX.	CS425	Digital Image Processing	2	3(3+0)/3(2+1)	

*As per HEC guidelines, a list of course options should be provided. However, for smooth administration of teaching and examinations (including affiliated colleges), only the courses listed in this table are offered. To further enhance students' knowledge, one additional course is proposed in the specialization, in addition to the eight courses suggested by HEC.

**The list of courses may be updated by the Chairman based on market needs or advancements in the domain.

2.10. SPECIALIZATION ELECTIVE COURSES (SOFTWARE)

Sp. #	Code	Course Title	Cat. Type	Credit Hrs	Pre-Req.
I.	CS331	Software Requirement Engineering	3	3(3+0)	
II.	<<	Machine Learning	4	3(3+0)/3(2+1)	
III.	CS333	Software Design and Architecture	3	3(3+0)	
IV.	CS334	Formal Methods and Design	3	3(3+0)	
V.	CS431	Human Computer Interaction >>	3	3(3+0)/3(2+1)	
VI.	<<	Deep Learning	4	3(3+0)/3(2+1)	
VII.	CS433	Software Project Management	3	3(3+0)	
VIII.	CS434	Parallel and Distributed Computing	3	3(3+0)/3(2+1)	
IX.	CS435	Software Quality and Testing	3	3(3+0)/3(2+1)	

*As per HEC guidelines, a list of course options should be provided. However, for smooth administration of teaching and examinations (including affiliated colleges), only the courses listed in this table are offered. To further enhance students' knowledge, one additional course is proposed in the specialization, in addition to the eight courses suggested by HEC.

**The list of courses may be updated by the Chairman based on market needs or advancements in the domain.

2.11. SPECIALIZATION ELECTIVE COURSES (ARTIFICIAL INTELLIGENCE)

Sp. #	Code	Course Title	Cat. Type	Credit Hrs	Pre-Req.
I.	CS341	Programming for AI	4	3(3+0)/3(2+1)	
II.	CS342	Machine Learning >>	4	3(3+0)/3(2+1)	
III.	CS343	Knowledge Representation and Reasoning	4	3(3+0)	
IV.	CS344	Computer Vision	4	3(3+0)/3(2+1)	

V.	<<	Human Computer Interaction	3	3(3+0)/3(2+1)	
VI.	CS442	Deep Learning >>	4	3(3+0)/3(2+1)	
VII.	CS443	Natural Language Processing	4	3(3+0)/3(2+1)	
VIII.	CS444	Evolutionary Computing & Swarm Intelligence	4	3(3+0)	
IX.	CS445	Reinforcement Learning	4	3(3+0)/3(2+1)	

*As per HEC guidelines, a list of course options should be provided. However, for smooth administration of teaching and examinations (including affiliated colleges), only the courses listed in this table are offered. To further enhance students' knowledge, one additional course is proposed in the specialization, in addition to the eight courses suggested by HEC.

**The list of courses may be updated by the Chairman based on market needs or advancements in the domain.

2.12. SPECIALIZATION ELECTIVE COURSES (CYBER SECURITY)

Sp. #	Code	Course Title	Cat. Type	Credit Hrs	Pre-Req.
I.	CS351	Wireless and Mobile Security	5	3(3+0)	
II.	<<	Machine Learning	2	3(3+0)/3(2+1)	
III.	CS353	Secure Software Design and Development	5	3(3+0)/3(2+1)	
IV.	CS354	Digital Forensics and Incident Response	5	3(3+0)	
V.	CS451	Network Security >>	5	3(3+0)	
VI.	CS452	Applied Cryptography	5	3(3+0)/3(2+1)	
VII.	CS453	Usable Security and Privacy	5	3(3+0)	
VIII.	CS454	Cloud Security	5	3(3+0)	
IX.	CS455	Artificial intelligence for Cyber Security	5	3(3+0)	

*As per HEC guidelines, a list of course options should be provided. However, for smooth administration of teaching and examinations (including affiliated colleges), only the courses listed in this table are offered. To further enhance students' knowledge, one additional course is proposed in the specialization, in addition to the eight courses suggested by HEC.

**The list of courses may be updated by the Chairman based on market needs or advancements in the domain.

2.13. ALL SPECIALIZATION COURSES IN A SEQUENCE

Sp. #	Honors	Artificial Intelligence	Software Eng.	Cyber Security
I.	Web Technologies	Programming for AI	Software Requirement Eng.	Wireless and Mobile Security
II.	Machine Learning	Machine Learning	Machine Learning	Machine Learning
III.	Advanced Programming*	Knowledge Repr. and Reasoning	Software Design and Architecture	Secure Software Design and Dev.
IV.	Web Engineering*	Computer Vision	Formal Methods and Design	Digital Forensics and Incident Resp.
V.	Network Security	Human Computer Interaction	Human Computer Interaction	Network Security
VI.	Deep Learning	Deep Learning	Deep Learning	Applied Cryptography
VII.	Mobile Application Development*	Natural Language Processing	Software Project Management	Usable Security and Privacy
VIII.	Internet of Things and Its App.	Evol. Com. & Swarm Intel.	Parallel and Distributed Comp.	Cloud Security
IX.	Digital Image Processing	Reinforcement Learning	Software Quality and Testing	Artificial Intel. for Cyber Security

*Common courses across different specializations are offered in the same semester to maintain a uniform schedule.

2.14. FYP EVALUATION CRITERIA

The following criteria are used to evaluate FYP I & II, comprising 200 marks.

S#	Component	Weightage	Marks	Rubrics
1.	Monthly Meetings	25%	50	- Attendance & punctuality (10) - Meeting preparedness (10) - Progress demonstrated (20) - Professional behavior & communication (10)
2.	External Viva	25%	50	- Understanding of full project (15) - Methodology + results (15)

				• Contribution/innovation (10) • Presentation + Q&A performance (10)
3.	1 st Internal Viva	12.5%	25	• Problem clarity (5) • Literature understanding (5) • Early methodology/framework (10) • Presentation quality (5)
4.	2 nd Internal Viva	12.5%	25	• Implementation progress (10) • Quality of partial results (10) • Preparedness for final stages (5)
5.	Thesis/Report	7.5%	15	• Structure & formatting (3) • Literature review depth (4) • Methodology clarity (3) • Findings & discussion (4) • Referencing style (1)
6.	Code/Practical Work	7.5%	15	• Functionality & accuracy (6) • Innovation or complexity (4) • Documentation (3) • Reproducibility (2)
7.	Similarity Report	5%	10	0–10%: 10 marks 11–20%: 7–8 marks 21–30%: 4–6 marks 31–40%: 1–3 marks >40%: 0 marks (revision required)
8.	Proposal Submission	5%	10	• Problem statement clarity (3) • Objectives & research questions (3) • Feasibility & methodology (3) • Formatting & submission on time (1)

*The Chairman is authorized to adjust the mark distribution/assignments.

2.15. INTERNSHIP EVALUATION CRITERIA

The following criteria are used to evaluate an internship, comprising 100 marks.

S#	Component	Weightage	Marks	Rubrics
1.	Site Supervisor Evaluations	40%	40	• Quality, accuracy, time of task (10) • Problem solving skills (10) • Skills in policies, time management, deadlines (5) • Communication (5) • Teamwork, cooperation, conflict resolution (5) • Punctuality, integrity, motivation (5)
2.	Student Reports	30%	30	• Completeness, relevance, clarity (10) • Understanding of tasks & learning outcomes (10) • Logical organization, formatting, referencing (5) • Grammar, coherence, professionalism (5)
3.	Activity log completion	5%	5	• Detailed and correct entries, • Clear description of tasks performed, • Submitted on schedule (5)
4.	Faculty supervisor Evaluation	25%	25	• Depth of learning and comprehension (10) • Interaction, responsiveness, commitment (5) • Reports, logs, communication (5) • Improvement over internship duration (5)

*The Chairman is authorized to adjust these assignments. Additionally, the chairman may modify the level or type of internship, whether conducted at the parent university or any other organization/office.

3. ASSOCIATE DEGREE - COMPUTING

3.1. GENERAL EDUCATION COURSES

Same as given in the BS Computer Science program.

S#	Code	Course Title	Sub-Category	Credit Hrs	Pre-Req
1	XX###	Applied Physics*	Natural Sciences	3(2+1)	

		<i>(Choose only ONE Course)</i> -Introductory Mechanics -Modern Physics -Introductory Electricity -Everyday Science			
2	XX###	Functional English	ENG	3(3+0)	
3	XX###	Expository Writing	ENG	3(3+0)	
4	XX###	Applied Mathematics for Computing I*	QR-I	3(3+0)	
5	XX###	Applied Mathematics for Computing II*	QR-II	3(3+0)	
6	XX###	Application of ICT	-	3(2+1)	
7	XX###	Introduction to Management*	Social Sciences	2(2+0)	
		<i>(Choose any ONE Course)</i> -Economics			
8	XX###	Professional Practices*	Arts & Humanities	2(2+0)	
		<i>(Choose any ONE Course)</i> -(Any other Arts & Humanities Course) Introduction to Journalism Introduction to Philosophy			
9	XX###	Entrepreneurship	-	2(2+0)	
10	XX###	Civics and Community Management	-	2(2+0)	
11	XX###	Pakistan Studies	-	2(2+0)	
12	XX###	Islamic Studies / Ethics	-	2(2+0)	
13	XX###	Ideology and Constitution of Pakistan	-	2(2+0)	
14	XX###	Fehm e Quran I	-	1(1+0)	
15	XX###	Fehm e Quran II	-	1(1+0)	

*As per HEC guidelines, there should be a list of options to select courses, but for the smooth administration of teaching and examination (including affiliated colleges), we are restricting that to only offering these courses.

3.2. FUNDAMENTAL COMPUTING OR MATHEMATICAL COURSES

S#	Code	Course Title	Cat. Type	Credit Hrs	Pre-Req.
1.	CS101	Applications of ICT	0	3 (2+1)	
2.	CS102	Applied Mathematics for Computing I	0	3 (3+0)	
3.	CS103	Applied Mathematics for Computing II	0	3 (3+0)	CS102

3.3. COMPUTING CORE COURSES

Students are required to take these compulsory computer science courses.

S#	Code	Course Title	Cat. Type	Credit Hrs	Pre-Req
1.	CS111	Programming Fundamentals	1	4 (3+1)	
2.	CS112	Object Oriented Programming	1	4 (3+1)	CS111
3.	CS211	Data Structures	1	4 (3+1)	CS111
4.	CS212	Database Systems	1	4 (3+1)	
5.	CS213	Software Engineering	1	3 (3+0)	
6.	CS215	Operating Systems	1	4 (3+1)	CS101

7.	CS216	Computer Networks	1	3 (3+0)	
----	-------	-------------------	---	---------	--

3.4. MANDATORY LAB COURSES

S#	Code	Course Title	Cat. Type	Credit Hrs	Co Pre-Req.
1.	CS111L	Programming Fundamentals	1	1 (0+3)	CS111
2.	CS112L	Object Oriented Programming	1	1 (0+3)	CS112
3.	CS211L	Data Structures	1	1 (0+3)	CS211
4.	CS212L	Database Systems	1	1 (0+3)	CS212
5.	CS215L	Operating Systems	1	1 (0+3)	CS215

3.5. POOL OF ELECTIVE COURSES

Students can select these courses as specialization electives.

S#	Code	Course Title	Cat. Type	Credit Hrs	Pre-Req
1.	CS113	Digital Logic Design	1	4 (3+1)	
2.	CS214	Design and Analysis of Algorithms	1	3 (3+0)	
3.	CS261	Web Development (Frontend and Backend)	6	3 (3+0)	
4.	CS262	Introduction to ML and AI Tools	6	3 (3+0)	
1.		Theory of Automata	3	3 (3+0)	
2.		Software Project Management	1	3 (3+0)	
3.		Mobile App Development	1	3 (3+0)	
4.		Data Analytics and SQL for Practitioners	2	3 (3+0)	
5.		Information Security	4	3 (3+0)	
6.		Cybersecurity Fundamentals	2	3 (3+0)	
7.		DevOps and CI/CD Essentials	3	3 (3+0)	
8.		Database Administration and SQL Tuning	3	3 (3+0)	
9.		UX/UI and Product Prototyping	3	3 (3+0)	
10.		IoT Basics	4	3 (3+0)	
11.		Python Scripting	4	3 (3+0)	
12.		Networking and Wireless	4	3 (3+0)	
13.		Application Development	5	3 (3+0)	
14.		Blockchain	5	3 (3+0)	
15.		Computer Organization and Architecture	2	3 (2+1)	
16.		Cloud Computing	4	3 (3+0)	

*As per HEC guidelines, a list of options should be provided to select 3–4 courses for 12 credit hours. However, for smooth administration of teaching and examinations (including affiliated colleges), only the courses listed here will be offered.

4. PROGRAM'S DETAILS

4.1. PROGRAM LEARNING OUTCOMES FOR BS COMPUTER SCIENCE

S#	PLOs	Computing Graduate
----	------	--------------------

1.	Academic Education	To prepare graduates as computing professionals
2.	Knowledge for Solving Computing Problems	Apply knowledge of computing fundamentals, knowledge of a computing specialization, and mathematics, science, and domain knowledge appropriate for the computing specialization to the abstraction and conceptualization of computing models from defined problems and requirements.
3.	Problem Analysis	Identify, formulate, research literature, and solve complex computing problems reaching substantiated conclusions using fundamental principles of mathematics, computing sciences, and relevant domain disciplines.
4.	Design/ Development of Solutions	Design and evaluate solutions for complex computing problems, and design and evaluate systems, components, or processes that meet specified needs with appropriate consideration for public health and safety, cultural, societal, and environmental considerations.
5.	Modern Tool Usage	Create, select, adapt and apply appropriate techniques, resources, and modern computing tools to complex computing activities, with an understanding of the limitations.
6.	Individual and Teamwork	Function effectively as an individual and as a member or leader in diverse teams and in multi-disciplinary settings.
7.	Communication	Communicate effectively with the computing community and with society at large about complex computing activities by being able to comprehend and write effective reports, design documentation, make effective presentations, and give and understand clear instructions.
8.	Computing Professionalism and Society	Understand and assess societal, health, safety, legal, and cultural issues within local and global contexts, and the consequential responsibilities relevant to professional computing practice
9.	Ethics	Understand and commit to professional ethics, responsibilities, and norms of professional computing practice
10.	Life-long Learning	Recognize the need, and have the ability, to engage in independent learning for continual development as a computing professional

4.2. STANDARD NOMENCLATURE

S#	Program	Years	NQF	Title of Degree
1.	BS Computer Science	4	Level 06	BS Computer Science (Honors) BS Computer Science (Software) BS Computer Science (Artificial Intelligence) BS Computer Science (Cyber Security)
2.	Associate Degree	2	Level 05	Associate Degree - Computing

4.3. ELIGIBILITY CRITERIA

S#	Program	School Certificate	Studied	Percentage
1.	BS Computer Science	HSSC/A-levels or an IBCC equivalent (min 12 years of school)	FSc Pre-Eng/Comp.Sc A-Level/Diploma FSc Pre-Medical/FA	50%
2.	Associate Degree Computing	HSSC/A-levels or an IBCC equivalent (min 12 years of school)	FSc Pre-Eng/Comp.Sc A-Level/Diploma FSc Pre-Medical/FA	50%*

*As per HEC guidelines, an Associate Degree may have a 45% passing requirement; however, we maintain a 50% passing requirement.

The criteria allow students who have not studied Mathematics at the intermediate level. However, they must pass deficiency courses in Mathematics (06 credit hours) during the first year, including the zero semester.

S#	Courses	HSSC (No Mathematics)	HSCC (Mathematics)	Semester
1.	Applied Mathematics for Computer I	Compulsory Math I / QR1	QR1	I
2.	Applied Mathematics for Computer II	Compulsory Math II / QR1I	QR1I	II

*As per HEC guidelines, students with HSSC (without Mathematics) must study these Mathematics courses.

**To strengthen the mathematical background, the department makes these courses mandatory for all students (with or without Mathematics) and counts them as QR courses in the General Education category.

4.4. ASSOCIATE DEGREE PATHWAY (ENTRY, EXIT, AND TRANSFER REQUIREMENTS)

4.4.1. Exit Option and Associate Degree Requirements:

Students enrolled in the BS Computer Science program may exit the program after completion of the 4th semester. To be awarded the Associate Degree in Computing, students must complete two elective courses in addition to all the courses distributed over the first four semesters in the BS program. These elective courses may be:

- The two elective courses specified in the semester wise plan for the Associate Degree, or
- Any other elective courses offered by the department that fulfill the elective course requirements of the Associate Degree Computing.

These elective courses may be completed during the 3rd or 4th semester or during the summer semester, as per the needs of the students.

4.4.2. Direct Entry to 5th Semester (Lateral Admission):

Only students possessing 14 years of education in Computing are eligible for direct admission into the 5th semester (commonly known as lateral admissions) of the BS Computer Science program.

a. Lateral Admission for Associate Degree Computing Holders:

Students holding an Associate Degree in Computing, with a curriculum structure similar to the BS Computer Science program implemented at AWKUM, are required to complete only IDS-I (Calculus & Analytical Geometry) during the summer semester. Upon successful completion, these students may join the 5th semester of the BS program without any semester gap.

b. Lateral Admission from Other Equivalent Programs

Students joining the BS Computer Science program from any other equivalent program must complete deficiency courses as identified from their academic transcripts. The deficiency requirements range from 15 to 18 credit hours, which must be completed through a bridging semester prior to enrollment in the 5th semester of the BS Computer Science program.

4.5. IMPLEMENTATION

- This scheme of studies is proposed to be implemented for students enrolling at the department starting from Fall 2026.
- It is retrospectively applied to students enrolled in Fall 2025 with the approval of the Board of Studies, in anticipation of the approval of the Board of Faculty/Academic Council.
- The Board of Studies also authorized the Chairman of the department to adjust the order of courses offered in different semesters as per the availability of subject experts and other resources.

4.6. COURSE DESCRIPTION

The course details along with their CLOs and the mapping of CLOs with PLOs (BT levels) are given in the Annexure given with this document. It also contains the recommended course contents, recommended material, and the week-wise distribution of topics.

Details about recommended professional certifications are provided.

5. SEMESTER WISE COURSE DISTRIBUTION

5.1. SEMESTER WISE COURSE DISTRIBUTION (BS COMPUTER SCIENCE – ALL DISCIPLINES)

Sem	Flag	Code	Course Title	CH	Pre-Req	Coverage															
Semester 1		CS111	Computer Sc. Core (Programming Fundamentals)	4(3+1)		<table><tr><th>Cat.</th><th>Courses</th><th>T. Credits</th></tr><tr><td> Comp. Core</td><td>1</td><td>4</td></tr><tr><td> GE</td><td>5</td><td>14</td></tr><tr><td>Total</td><td>6</td><td>18</td></tr></table>	Cat.	Courses	T. Credits	 Comp. Core	1	4	 GE	5	14	Total	6	18			
	Cat.	Courses	T. Credits																		
	 Comp. Core	1	4																		
	 GE	5	14																		
	Total	6	18																		
		ENG100	Gen. Edu. (Functional English)	3(3+0)																	
	CS101	Gen. Edu. (Applications of ICT)	3(2+1)																		
	PHY102	Gen. Edu. (NS - Applied Physics)	3(2+1)																		
	CS102	Gen. Edu. (QR1 - AMC I)	3(3+0)																		
	MAN101	Gen. Edu. (SS – Introduction to Management)	2(2+0)																		
Semester 2		CS112	Comp. Sc. Core (Object Oriented Programming)	4(3+1)	CS111	<table><tr><th>Cat.</th><th>Courses</th><th>T. Credits</th></tr><tr><td> Comp. Core</td><td>2</td><td>8</td></tr><tr><td> GE</td><td>4</td><td>10</td></tr><tr><td>Total</td><td>6</td><td>18</td></tr></table>	Cat.	Courses	T. Credits	 Comp. Core	2	8	 GE	4	10	Total	6	18			
	Cat.	Courses	T. Credits																		
	 Comp. Core	2	8																		
	 GE	4	10																		
	Total	6	18																		
		CS113	Computer Sc. Core (Digital Logic Design)	4(3+1)																	
	ENG200	Gen. Edu. (Expository Writing)	3(3+0)																		
	CS103	Gen. Edu. (QR II - AMC II)	3(3+0)	CS102																	
	ISL100	Gen. Edu. (Islamic Studies / Ethics – ETH100)	2(2+0)																		
	PKS100	Gen. Edu. (Pakistan Studies)	2(2+0)																		
Semester 3		CS211	Computer Sc. Core (Data Structures)	4(3+1)	CS111	<table><tr><th>Cat.</th><th>Courses</th><th>T. Credits</th></tr><tr><td> Comp. Core</td><td>2</td><td>8</td></tr><tr><td> IDS</td><td>1</td><td>3</td></tr><tr><td> GE</td><td>4</td><td>7</td></tr><tr><td>Total</td><td>7</td><td>18</td></tr></table>	Cat.	Courses	T. Credits	 Comp. Core	2	8	 IDS	1	3	 GE	4	7	Total	7	18
	Cat.	Courses	T. Credits																		
	 Comp. Core	2	8																		
	 IDS	1	3																		
	 GE	4	7																		
	Total	7	18																		
		CS212	Computer Sc. Core (Database Systems)	4(3+1)																	
	MATH101	IDS I (Calculus & Analytical Geometry)	3(3+0)																		
	PKS200	Gen. Edu. (Ideology and Constitution of Pakistan)	2(2+0)																		
	MGT200	Gen. Edu. (AH – Professional Practices)	2(2+0)																		
	SWS200	Gen. Edu. (Civics and Community Engagement)	2(2+0)																		
	ISL200	Gen. Edu. (Fehm-e-Quran - I)	1(0+1)																		
Semester 4		CS215	Computer Sc. Core (Operating Systems)	4(3+1)	CS101	<table><tr><th>Cat.</th><th>Courses</th><th>T. Credits</th></tr><tr><td> Comp. Core</td><td>4</td><td>13</td></tr><tr><td> GE</td><td>2</td><td>3</td></tr><tr><td>Total</td><td>2</td><td>16</td></tr></table>	Cat.	Courses	T. Credits	 Comp. Core	4	13	 GE	2	3	Total	2	16			
	Cat.	Courses	T. Credits																		
	 Comp. Core	4	13																		
	 GE	2	3																		
	Total	2	16																		
		CS213	Computer Sc. Core (Software Engineering)	3(3+0)																	
	CS216	Computer Sc. Core (Computer Networks)	3(3+0)																		
	CS214	Comp. Sc. Core (Design & Analysis of Algorithm)	3(3+0)																		
	MGT400	Gen. Edu. (Entrepreneurship)	2(2+0)																		
	ISL300	Gen. Edu. (Fehm-e-Quran II)	1(0+1)																		
Semester 5		CS311	Computer Sc. Core (Artificial Intelligence)	3(3+0)	CS101	<table><tr><th>Cat.</th><th>Courses</th><th>T. Credits</th></tr><tr><td> Comp. Core</td><td>2</td><td>6</td></tr><tr><td> Sp. Elective</td><td>2</td><td>6</td></tr><tr><td> IDS</td><td>2</td><td>6</td></tr><tr><td>Total</td><td>6</td><td>18</td></tr></table>	Cat.	Courses	T. Credits	 Comp. Core	2	6	 Sp. Elective	2	6	 IDS	2	6	Total	6	18
	Cat.	Courses	T. Credits																		
	 Comp. Core	2	6																		
	 Sp. Elective	2	6																		
	 IDS	2	6																		
	Total	6	18																		
	CS312	Computer Sc. Core (Theory of Automata)	3(3+0)	CS103																	
		Specialization Elective I	3(3+0)																		
		Specialization Elective II	3(3+0)																		
	MATH111	IDS II (Linear Algebra)	3(3+0)																		
	MATH198	IDS III (Discrete Structure)	3(3+0)																		
Semester 6		CS411	Computer Sc. Core (Information Security)	3(3+0)	CS103	<table><tr><th>Cat.</th><th>Courses</th><th>T. Credits</th></tr><tr><td> Comp. Core</td><td>2</td><td>6</td></tr><tr><td> Sp. Elective</td><td>2</td><td>6</td></tr><tr><td> IDS</td><td>1</td><td>3</td></tr><tr><td>Total</td><td>5</td><td>15</td></tr></table>	Cat.	Courses	T. Credits	 Comp. Core	2	6	 Sp. Elective	2	6	 IDS	1	3	Total	5	15
	Cat.	Courses	T. Credits																		
	 Comp. Core	2	6																		
	 Sp. Elective	2	6																		
	 IDS	1	3																		
	Total	5	15																		
	CS412	Comp. Sc. Core (Computer Org & Architecture)	3(2+1)	CS113																	
		Specialization Elective III	3(3+0)																		
		Specialization Elective IV	3(3+0)																		
	STAT300	IDS IV (Probability & Statistics)	3(3+0)																		
Semester 7		CS413	Computer Sc. Core (Cloud Computing)	3(3+0)	CS101	<table><tr><th>Cat.</th><th>Courses</th><th>T. Credits</th></tr><tr><td> Comp. Core</td><td>1</td><td>3</td></tr><tr><td> Sp. Elective</td><td>2</td><td>6</td></tr><tr><td> Professional</td><td>2</td><td>6</td></tr><tr><td>Total</td><td>5</td><td>15</td></tr></table>	Cat.	Courses	T. Credits	 Comp. Core	1	3	 Sp. Elective	2	6	 Professional	2	6	Total	5	15
	Cat.	Courses	T. Credits																		
	 Comp. Core	1	3																		
	 Sp. Elective	2	6																		
	 Professional	2	6																		
	Total	5	15																		
		Specialization Elective V	3(3+0)																		
		Specialization Elective VI	3(3+0)																		
	CS497	Field Experience (Internship)	3(3+0)																		
	CS498	Computer Sc. Core (Final Year Project I)	3(3+0)																		
Semester 8			Specialization Elective VII	3(3+0)	<table><tr><th>Cat.</th><th>Courses</th><th>T. Credits</th></tr><tr><td> Sp. Elective</td><td>3</td><td>9</td></tr><tr><td> Professional</td><td>2</td><td>6</td></tr><tr><td>Total</td><td>5</td><td>15</td></tr></table>	Cat.	Courses	T. Credits	 Sp. Elective	3	9	 Professional	2	6	Total	5	15				
	Cat.	Courses	T. Credits																		
	 Sp. Elective	3	9																		
	 Professional	2	6																		
	Total	5	15																		
			Specialization Elective VIII	3(3+0)																	
		Specialization Elective IX	3(3+0)																		
	CS496	Professional Certification	3(3+0)																		
	CS499	Computer Sc. Core (Final Year Project II)	3(3+0)																		

Total 133 Credit Hours

*General Education courses are scheduled within the first four semesters. Internship (3 CH) must be completed as per HEC policy and may be offered during the summer break after Semester 6. Students must complete compulsory professional certification(s) equivalent to 3 credit hours (16 study hours = 1 credit hour). Final Year Project is offered as FYP-I and FYP-II, each evaluated independently with 100 marks. Applied Mathematics for Computing I and II (Math-I and Math-II) are mandatory for all students and are offered to meet the mathematics requirements for those without Mathematics at HSSC. The Chairman may change the order of subjects based on the availability of faculty and resources. Course code may be changed subject to the HEC/ERP system configuration

5.2. SEMESTER WISE COURSE DISTRIBUTION (BS COMPUTER SCIENCE - HONORS)

Sem	Flag	Code	Course Title	CH	Pre-Req	Coverage																	
Semester 1		CS111	Computer Sc. Core (Programming Fundamentals)	4(3+1)		<table><tr><th>Cat.</th><th>Courses</th><th>T. Credits</th></tr><tr><td> Comp. Core</td><td>1</td><td>4</td></tr><tr><td> GE</td><td>5</td><td>14</td></tr><tr><td>Total</td><td>6</td><td>18</td></tr></table>			Cat.	Courses	T. Credits	 Comp. Core	1	4	 GE	5	14	Total	6	18			
	Cat.	Courses	T. Credits																				
	 Comp. Core	1	4																				
	 GE	5	14																				
	Total	6	18																				
		ENG100	Gen. Edu. (Functional English)	3(3+0)																			
	CS101	Gen. Edu. (Applications of ICT)	3(2+1)																				
	PHY102	Gen. Edu. (NS - Applied Physics)	3(2+1)																				
	CS102	Gen. Edu. (QR1 - AMC I)	3(3+0)																				
	MAN101	Gen. Edu. (SS – Introduction to Management)	2(2+0)																				
Semester 2		CS112	Comp. Sc. Core (Object Oriented Programming)	4(3+1)	CS111	<table><tr><th>Cat.</th><th>Courses</th><th>T. Credits</th></tr><tr><td> Comp. Core</td><td>2</td><td>8</td></tr><tr><td> GE</td><td>4</td><td>10</td></tr><tr><td>Total</td><td>6</td><td>18</td></tr></table>			Cat.	Courses	T. Credits	 Comp. Core	2	8	 GE	4	10	Total	6	18			
	Cat.	Courses	T. Credits																				
	 Comp. Core	2	8																				
	 GE	4	10																				
	Total	6	18																				
		CS113	Computer Sc. Core (Digital Logic Design)	4(3+1)																			
	ENG200	Gen. Edu. (Expository Writing)	3(3+0)																				
	CS103	Gen. Edu. (QR II - AMC II)	3(3+0)	CS102																			
	ISL100	Gen. Edu. (Islamic Studies / Ethics)	2(2+0)																				
	PKS100	Gen. Edu. (Pakistan Studies)	2(2+0)																				
Semester 3		CS211	Computer Sc. Core (Data Structures)	4(3+1)	CS111	<table><tr><th>Cat.</th><th>Courses</th><th>T. Credits</th></tr><tr><td> Comp. Core</td><td>2</td><td>8</td></tr><tr><td> IDS</td><td>1</td><td>3</td></tr><tr><td> GE</td><td>4</td><td>7</td></tr><tr><td>Total</td><td>7</td><td>18</td></tr></table>			Cat.	Courses	T. Credits	 Comp. Core	2	8	 IDS	1	3	 GE	4	7	Total	7	18
	Cat.	Courses	T. Credits																				
	 Comp. Core	2	8																				
	 IDS	1	3																				
	 GE	4	7																				
	Total	7	18																				
		CS212	Computer Sc. Core (Database Systems)	4(3+1)																			
	MATH101	IDS I (Calculus & Analytical Geometry)	3(3+0)																				
	PKS200	Gen. Edu. (Ideology and Constitution of Pakistan)	2(2+0)																				
	MGT200	Gen. Edu. (AH – Professional Practices)	2(2+0)																				
	SWS200	Gen. Edu. (Civics and Community Engagement)	2(2+0)																				
	ISL200	Gen. Edu. (Fehm-e-Quran - I)	1(0+1)																				
Semester 4		CS215	Computer Sc. Core (Operating Systems)	4(3+1)	CS101	<table><tr><th>Cat.</th><th>Courses</th><th>T. Credits</th></tr><tr><td> Comp. Core</td><td>4</td><td>13</td></tr><tr><td> GE</td><td>2</td><td>3</td></tr><tr><td>Total</td><td>2</td><td>16</td></tr></table>			Cat.	Courses	T. Credits	 Comp. Core	4	13	 GE	2	3	Total	2	16			
	Cat.	Courses	T. Credits																				
	 Comp. Core	4	13																				
	 GE	2	3																				
	Total	2	16																				
		CS213	Computer Sc. Core (Software Engineering)	3(3+0)																			
	CS216	Computer Sc. Core (Computer Networks)	3(3+0)																				
	CS214	Comp. Sc. Core (Design & Analysis of Algorithm)	3(3+0)																				
	MGT400	Gen. Edu. (Entrepreneurship)	2(2+0)																				
	ISL300	Gen. Edu. (Fehm-e-Quran II)	1(0+1)																				
Semester 5		CS311	Computer Sc. Core (Artificial Intelligence)	3(3+0)	CS101	<table><tr><th>Cat.</th><th>Courses</th><th>T. Credits</th></tr><tr><td> Comp. Core</td><td>2</td><td>6</td></tr><tr><td> Sp. Elective</td><td>2</td><td>6</td></tr><tr><td> IDS</td><td>2</td><td>6</td></tr><tr><td>Total</td><td>6</td><td>18</td></tr></table>			Cat.	Courses	T. Credits	 Comp. Core	2	6	 Sp. Elective	2	6	 IDS	2	6	Total	6	18
	Cat.	Courses	T. Credits																				
	 Comp. Core	2	6																				
	 Sp. Elective	2	6																				
	 IDS	2	6																				
	Total	6	18																				
		CS312	Computer Sc. Core (Theory of Automata)	3(3+0)	CS103																		
	CS321	Sp. Elec. I (Web Technologies)	3(3+0)																				
	CS342	Sp. Elec. II (Machine Learning)	3(3+0)																				
	MATH111	IDS II (Linear Algebra)	3(3+0)																				
	MATH198	IDS III (Discrete Structure)	3(3+0)																				
Semester 6		CS411	Computer Sc. Core (Information Security)	3(3+0)	CS103	<table><tr><th>Cat.</th><th>Courses</th><th>T. Credits</th></tr><tr><td> Comp. Core</td><td>2</td><td>6</td></tr><tr><td> Sp. Elective</td><td>2</td><td>6</td></tr><tr><td> IDS</td><td>1</td><td>3</td></tr><tr><td>Total</td><td>5</td><td>15</td></tr></table>			Cat.	Courses	T. Credits	 Comp. Core	2	6	 Sp. Elective	2	6	 IDS	1	3	Total	5	15
	Cat.	Courses	T. Credits																				
	 Comp. Core	2	6																				
	 Sp. Elective	2	6																				
	 IDS	1	3																				
	Total	5	15																				
		CS412	Comp. Sc. Core (Computer Org & Architecture)	3(2+1)	CS113																		
	CS323	Sp. Elec. III (Advanced Programming)	3(3+0)																				
	CS324	Sp. Elec. IV (Web Engineering)	3(3+0)																				
	STAT300	IDS IV (Probability & Statistics)	3(3+0)																				
Semester 7		CS413	Computer Sc. Core (Cloud Computing)	3(3+0)	CS101	<table><tr><th>Cat.</th><th>Courses</th><th>T. Credits</th></tr><tr><td> Comp. Core</td><td>1</td><td>3</td></tr><tr><td> Sp. Elective</td><td>2</td><td>6</td></tr><tr><td> Professional</td><td>2</td><td>6</td></tr><tr><td>Total</td><td>5</td><td>15</td></tr></table>			Cat.	Courses	T. Credits	 Comp. Core	1	3	 Sp. Elective	2	6	 Professional	2	6	Total	5	15
	Cat.	Courses	T. Credits																				
	 Comp. Core	1	3																				
	 Sp. Elective	2	6																				
	 Professional	2	6																				
	Total	5	15																				
	CS451	Sp. Elec. V (Network Security)	3(3+0)																				
	CS442	Sp. Elec. VI (Deep Learning)	3(3+0)																				
	CS497	Field Experience (Internship)	3(3+0)																				
	CS498	Computer Sc. Core (Final Year Project I)	3(3+0)																				
Semester 8		CS423	Sp. Elec. VII (Mobile App. Development)	3(3+0)		<table><tr><th>Cat.</th><th>Courses</th><th>T. Credits</th></tr><tr><td> Sp. Elective</td><td>3</td><td>9</td></tr><tr><td> Professional</td><td>2</td><td>6</td></tr><tr><td>Total</td><td>5</td><td>15</td></tr></table>			Cat.	Courses	T. Credits	 Sp. Elective	3	9	 Professional	2	6	Total	5	15			
	Cat.	Courses	T. Credits																				
	 Sp. Elective	3	9																				
	 Professional	2	6																				
	Total	5	15																				
	CS424	Sp. Elec. VIII (Internet of Things and its Applications)	3(3+0)																				
	CS425	Sp. Elec. IX (Digital Image Processing)	3(3+0)																				
	CS496	Professional Certification	3(3+0)																				
	CS499	Computer Sc. Core (Final Year Project II)	3(3+0)																				
Total 133 Credit Hours																							

5.3. SEMESTER WISE COURSE DISTRIBUTION (BSCS – SOFTWARE)

Sem	Flag	Code	Course Title	CH	Pre-Req	Coverage																	
Semester 1		CS111	Comp. Sc. Core (Programming Fundamentals)	4(3+1)		<table><tr><th>Cat.</th><th>Courses</th><th>T. Credits</th></tr><tr><td> Comp. Core</td><td>1</td><td>4</td></tr><tr><td> GE</td><td>5</td><td>14</td></tr><tr><td>Total</td><td>6</td><td>18</td></tr></table>			Cat.	Courses	T. Credits	 Comp. Core	1	4	 GE	5	14	Total	6	18			
	Cat.	Courses	T. Credits																				
	 Comp. Core	1	4																				
	 GE	5	14																				
	Total	6	18																				
		ENG100	Gen. Edu. (Functional English)	3(3+0)																			
	CS101	Gen. Edu. (Applications of ICT)	3(2+1)																				
	PHY102	Gen. Edu. (NS - Applied Physics)	3(2+1)																				
	CS102	Gen. Edu. (QR1 - AMC I)	3(3+0)																				
	MAN101	Gen. Edu. (SS – Introduction to Management)	2(2+0)																				
Semester 2		CS112	Comp. Sc. Core (Object Oriented Programming)	4(3+1)	CS111	<table><tr><th>Cat.</th><th>Courses</th><th>T. Credits</th></tr><tr><td> Comp. Core</td><td>2</td><td>8</td></tr><tr><td> GE</td><td>4</td><td>10</td></tr><tr><td>Total</td><td>6</td><td>18</td></tr></table>			Cat.	Courses	T. Credits	 Comp. Core	2	8	 GE	4	10	Total	6	18			
	Cat.	Courses	T. Credits																				
	 Comp. Core	2	8																				
	 GE	4	10																				
	Total	6	18																				
		CS113	Computer Sc. Core (Digital Logic Design)	4(3+1)																			
	ENG200	Gen. Edu. (Expository Writing)	3(3+0)																				
	CS103	Gen. Edu. (QR II - AMC II)	3(3+0)	CS102																			
	ISL100	Gen. Edu. (Islamic Studies / Ethics)	2(2+0)																				
	PKS100	Gen. Edu. (Pakistan Studies)	2(2+0)																				
Semester 3		CS211	Computer Sc. Core (Data Structures)	4(3+1)	CS111	<table><tr><th>Cat.</th><th>Courses</th><th>T. Credits</th></tr><tr><td> Comp. Core</td><td>2</td><td>8</td></tr><tr><td> IDS</td><td>1</td><td>3</td></tr><tr><td> GE</td><td>4</td><td>7</td></tr><tr><td>Total</td><td>7</td><td>18</td></tr></table>			Cat.	Courses	T. Credits	 Comp. Core	2	8	 IDS	1	3	 GE	4	7	Total	7	18
	Cat.	Courses	T. Credits																				
	 Comp. Core	2	8																				
	 IDS	1	3																				
	 GE	4	7																				
	Total	7	18																				
		CS212	Computer Sc. Core (Database Systems)	4(3+1)																			
	MATH101	IDS I (Calculus & Analytical Geometry)	3(3+0)																				
	PKS200	Gen. Edu. (Ideology and Constitution of Pakistan)	2(2+0)																				
	MGT200	Gen. Edu. (AH – Professional Practices)	2(2+0)																				
	SWS200	Gen. Edu. (Civics and Community Engagement)	2(2+0)																				
	ISL200	Gen. Edu. (Fehm-e-Quran - I)	1(0+1)																				
Semester 4		CS215	Computer Sc. Core (Operating Systems)	4(3+1)	CS101	<table><tr><th>Cat.</th><th>Courses</th><th>T. Credits</th></tr><tr><td> Comp. Core</td><td>4</td><td>13</td></tr><tr><td> GE</td><td>2</td><td>3</td></tr><tr><td>Total</td><td>2</td><td>16</td></tr></table>			Cat.	Courses	T. Credits	 Comp. Core	4	13	 GE	2	3	Total	2	16			
	Cat.	Courses	T. Credits																				
	 Comp. Core	4	13																				
	 GE	2	3																				
	Total	2	16																				
		CS213	Computer Sc. Core (Software Engineering)	3(3+0)																			
	CS216	Computer Sc. Core (Computer Networks)	3(3+0)																				
	CS214	Comp. Sc. Core (Design & Analysis of Algorithm)	3(3+0)																				
	MGT400	Gen. Edu. (Entrepreneurship)	2(2+0)																				
	ISL300	Gen. Edu. (Fehm-e-Quran II)	1(0+1)																				
Semester 5		CS311	Computer Sc. Core (Artificial Intelligence)	3(3+0)	CS101	<table><tr><th>Cat.</th><th>Courses</th><th>T. Credits</th></tr><tr><td> Comp. Core</td><td>2</td><td>6</td></tr><tr><td> Sp. Elective</td><td>2</td><td>6</td></tr><tr><td> IDS</td><td>2</td><td>6</td></tr><tr><td>Total</td><td>6</td><td>18</td></tr></table>			Cat.	Courses	T. Credits	 Comp. Core	2	6	 Sp. Elective	2	6	 IDS	2	6	Total	6	18
	Cat.	Courses	T. Credits																				
	 Comp. Core	2	6																				
	 Sp. Elective	2	6																				
	 IDS	2	6																				
	Total	6	18																				
	CS312	Computer Sc. Core (Theory of Automata)	3(3+0)	CS103																			
	CS331	Sp. Elec. I (Software Requirement Engineering)	3(3+0)																				
	CS342	Sp. Elec. II (Machine Learning)	3(3+0)																				
	MATH111	IDS II (Linear Algebra)	3(3+0)																				
	MATH198	IDS III (Discrete Structure)	3(3+0)																				
Semester 6		CS411	Computer Sc. Core (Information Security)	3(3+0)	CS103	<table><tr><th>Cat.</th><th>Courses</th><th>T. Credits</th></tr><tr><td> Comp. Core</td><td>2</td><td>6</td></tr><tr><td> Sp. Elective</td><td>2</td><td>6</td></tr><tr><td> IDS</td><td>1</td><td>3</td></tr><tr><td>Total</td><td>5</td><td>15</td></tr></table>			Cat.	Courses	T. Credits	 Comp. Core	2	6	 Sp. Elective	2	6	 IDS	1	3	Total	5	15
	Cat.	Courses	T. Credits																				
	 Comp. Core	2	6																				
	 Sp. Elective	2	6																				
	 IDS	1	3																				
	Total	5	15																				
	CS412	Comp. Sc. Core (Computer Org & Architecture)	3(2+1)	CS113																			
	CS333	Sp. Elec. III (Software Design and Architecture)	3(3+0)																				
	CS334	Sp. Elec. IV (Formal Methods and Design)	3(3+0)																				
	STAT300	IDS IV (Probability & Statistics)	3(3+0)																				
Semester 7		CS413	Computer Sc. Core (Cloud Computing)	3(3+0)	CS101	<table><tr><th>Cat.</th><th>Courses</th><th>T. Credits</th></tr><tr><td> Comp. Core</td><td>1</td><td>3</td></tr><tr><td> Sp. Elective</td><td>2</td><td>6</td></tr><tr><td> Professional</td><td>2</td><td>6</td></tr><tr><td>Total</td><td>5</td><td>15</td></tr></table>			Cat.	Courses	T. Credits	 Comp. Core	1	3	 Sp. Elective	2	6	 Professional	2	6	Total	5	15
	Cat.	Courses	T. Credits																				
	 Comp. Core	1	3																				
	 Sp. Elective	2	6																				
	 Professional	2	6																				
	Total	5	15																				
	CS431	Sp. Elec. V (Human Computer Interaction)	3(3+0)																				
	CS442	Sp. Elec. VI (Deep Learning)	3(3+0)																				
	CS497	Field Experience (Internship)	3(3+0)																				
	CS498	Computer Sc. Core (Final Year Project I)	3(3+0)																				
Semester 8		CS433	Sp. Elec. VII (Software Project Management)	3(3+0)		<table><tr><th>Cat.</th><th>Courses</th><th>T. Credits</th></tr><tr><td> Sp. Elective</td><td>3</td><td>9</td></tr><tr><td> Professional</td><td>2</td><td>6</td></tr><tr><td>Total</td><td>5</td><td>15</td></tr></table>			Cat.	Courses	T. Credits	 Sp. Elective	3	9	 Professional	2	6	Total	5	15			
	Cat.	Courses	T. Credits																				
	 Sp. Elective	3	9																				
	 Professional	2	6																				
	Total	5	15																				
	CS434	Sp. Elec. VIII (Parallel and Distributed Computing)	3(3+0)																				
	CS435	Sp. Elec. IX (Software Quality and Testing)	3(3+0)																				
	CS496	Professional Certification	3(3+0)																				
	CS499	Computer Sc. Core (Final Year Project II)	3(3+0)																				
Total 133 Credit Hours																							

5.4. SEMESTER WISE COURSE DISTRIBUTION (BSCS – ARTIFICIAL INTELLIGENCE)

Sem	Flag	Code	Course Title	CH	Pre-Req	Coverage															
Semester 1		CS111	Computer Sc. Core (Programming Fundamentals)	4(3+1)		<table><tr><th>Cat.</th><th>Courses</th><th>T. Credits</th></tr><tr><td> Comp. Core</td><td>1</td><td>4</td></tr><tr><td> GE</td><td>5</td><td>14</td></tr><tr><td>Total</td><td>6</td><td>18</td></tr></table>	Cat.	Courses	T. Credits	 Comp. Core	1	4	 GE	5	14	Total	6	18			
	Cat.	Courses	T. Credits																		
	 Comp. Core	1	4																		
	 GE	5	14																		
	Total	6	18																		
		ENG100	Gen. Edu. (Functional English)	3(3+0)																	
	CS101	Gen. Edu. (Applications of ICT)	3(2+1)																		
	PHY102	Gen. Edu. (NS - Applied Physics)	3(2+1)																		
	CS102	Gen. Edu. (QR1 - AMC I)	3(3+0)																		
	MAN101	Gen. Edu. (SS – Introduction to Management)	2(2+0)																		
Semester 2		CS112	Computer Sc. Core (Object Oriented Programming)	4(3+1)	CS111	<table><tr><th>Cat.</th><th>Courses</th><th>T. Credits</th></tr><tr><td> Comp. Core</td><td>2</td><td>8</td></tr><tr><td> GE</td><td>4</td><td>10</td></tr><tr><td>Total</td><td>6</td><td>18</td></tr></table>	Cat.	Courses	T. Credits	 Comp. Core	2	8	 GE	4	10	Total	6	18			
	Cat.	Courses	T. Credits																		
	 Comp. Core	2	8																		
	 GE	4	10																		
	Total	6	18																		
		CS113	Computer Sc. Core (Digital Logic Design)	4(3+1)																	
	ENG200	Gen. Edu. (Expository Writing)	3(3+0)																		
	CS103	Gen. Edu. (QR II - AMC II)	3(3+0)	CS102																	
	ISL100	Gen. Edu. (Islamic Studies / Ethics)	2(2+0)																		
	PKS100	Gen. Edu. (Pakistan Studies)	2(2+0)																		
Semester 3		CS211	Computer Sc. Core (Data Structures)	4(3+1)	CS111	<table><tr><th>Cat.</th><th>Courses</th><th>T. Credits</th></tr><tr><td> Comp. Core</td><td>2</td><td>8</td></tr><tr><td> IDS</td><td>1</td><td>3</td></tr><tr><td> GE</td><td>4</td><td>7</td></tr><tr><td>Total</td><td>7</td><td>18</td></tr></table>	Cat.	Courses	T. Credits	 Comp. Core	2	8	 IDS	1	3	 GE	4	7	Total	7	18
	Cat.	Courses	T. Credits																		
	 Comp. Core	2	8																		
	 IDS	1	3																		
	 GE	4	7																		
	Total	7	18																		
		CS212	Computer Sc. Core (Database Systems)	4(3+1)																	
	MATH101	IDS I (Calculus & Analytical Geometry)	3(3+0)																		
	PKS200	Gen. Edu. (Ideology and Constitution of Pakistan)	2(2+0)																		
	MGT200	Gen. Edu. (AH – Professional Practices)	2(2+0)																		
	SWS200	Gen. Edu. (Civics and Community Engagement)	2(2+0)																		
	ISL200	Gen. Edu. (Fehm-e-Quran - I)	1(0+1)																		
Semester 4		CS215	Computer Sc. Core (Operating Systems)	4(3+1)	CS101	<table><tr><th>Cat.</th><th>Courses</th><th>T. Credits</th></tr><tr><td> Comp. Core</td><td>4</td><td>13</td></tr><tr><td> GE</td><td>2</td><td>3</td></tr><tr><td>Total</td><td>2</td><td>16</td></tr></table>	Cat.	Courses	T. Credits	 Comp. Core	4	13	 GE	2	3	Total	2	16			
	Cat.	Courses	T. Credits																		
	 Comp. Core	4	13																		
	 GE	2	3																		
	Total	2	16																		
		CS213	Computer Sc. Core (Software Engineering)	3(3+0)																	
	CS216	Computer Sc. Core (Computer Networks)	3(3+0)																		
	CS214	Computer Sc. Core (Design & Analysis of Algorithm)	3(3+0)																		
	MGT400	Gen. Edu. (Entrepreneurship)	2(2+0)																		
	ISL300	Gen. Edu. (Fehm-e-Quran II)	1(0+1)																		
Semester 5		CS311	Computer Sc. Core (Artificial Intelligence)	3(3+0)	CS101	<table><tr><th>Cat.</th><th>Courses</th><th>T. Credits</th></tr><tr><td> Comp. Core</td><td>2</td><td>6</td></tr><tr><td> Sp. Elective</td><td>2</td><td>6</td></tr><tr><td> IDS</td><td>2</td><td>6</td></tr><tr><td>Total</td><td>6</td><td>18</td></tr></table>	Cat.	Courses	T. Credits	 Comp. Core	2	6	 Sp. Elective	2	6	 IDS	2	6	Total	6	18
	Cat.	Courses	T. Credits																		
	 Comp. Core	2	6																		
	 Sp. Elective	2	6																		
	 IDS	2	6																		
	Total	6	18																		
	CS312	Computer Sc. Core (Theory of Automata)	3(3+0)	CS103																	
	CS341	Sp. Elec. I (Programming for AI)	3(3+0)																		
	CS342	Sp. Elec. II (Machine Learning)	3(3+0)																		
	MATH111	IDS II (Linear Algebra)	3(3+0)																		
	MATH198	IDS III (Discrete Structure)	3(3+0)																		
Semester 6		CS411	Computer Sc. Core (Information Security)	3(3+0)	CS103	<table><tr><th>Cat.</th><th>Courses</th><th>T. Credits</th></tr><tr><td> Comp. Core</td><td>2</td><td>6</td></tr><tr><td> Sp. Elective</td><td>2</td><td>6</td></tr><tr><td> IDS</td><td>1</td><td>3</td></tr><tr><td>Total</td><td>5</td><td>15</td></tr></table>	Cat.	Courses	T. Credits	 Comp. Core	2	6	 Sp. Elective	2	6	 IDS	1	3	Total	5	15
	Cat.	Courses	T. Credits																		
	 Comp. Core	2	6																		
	 Sp. Elective	2	6																		
	 IDS	1	3																		
	Total	5	15																		
	CS412	Computer Sc. Core (Computer Org & Architecture)	3(2+1)	CS113																	
	CS343	Sp. Elec. III (Knowledge Rep. and Reasoning)	3(3+0)																		
	CS344	Sp. Elec. IV (Computer Vision)	3(3+0)																		
	STAT300	IDS IV (Probability & Statistics)	3(3+0)																		
Semester 7		CS413	Computer Sc. Core (Cloud Computing)	3(3+0)	CS101	<table><tr><th>Cat.</th><th>Courses</th><th>T. Credits</th></tr><tr><td> Comp. Core</td><td>1</td><td>3</td></tr><tr><td> Sp. Elective</td><td>2</td><td>6</td></tr><tr><td> Professional</td><td>2</td><td>6</td></tr><tr><td>Total</td><td>5</td><td>15</td></tr></table>	Cat.	Courses	T. Credits	 Comp. Core	1	3	 Sp. Elective	2	6	 Professional	2	6	Total	5	15
	Cat.	Courses	T. Credits																		
	 Comp. Core	1	3																		
	 Sp. Elective	2	6																		
	 Professional	2	6																		
	Total	5	15																		
	CS431	Sp. Elec. V (Human Computer Interaction)	3(3+0)																		
	CS442	Sp. Elec. VI (Deep Learning)	3(3+0)																		
	CS497	Field Experience (Internship)	3(3+0)																		
	CS498	Computer Sc. Core (Final Year Project I)	3(3+0)																		
Semester 8		CS443	Sp. Elec. VII (Natural Language Processing)	3(3+0)		<table><tr><th>Cat.</th><th>Courses</th><th>T. Credits</th></tr><tr><td> Sp. Elective</td><td>3</td><td>9</td></tr><tr><td> Professional</td><td>2</td><td>6</td></tr><tr><td>Total</td><td>5</td><td>15</td></tr></table>	Cat.	Courses	T. Credits	 Sp. Elective	3	9	 Professional	2	6	Total	5	15			
	Cat.	Courses	T. Credits																		
	 Sp. Elective	3	9																		
	 Professional	2	6																		
	Total	5	15																		
	CS444	Sp. Elec. VIII (Evolutionary Comp & Swarm Intel.)	3(3+0)																		
	CS445	Sp. Elec. IX (Reinforcement Learning)	3(3+0)																		
	CS496	Professional Certification	3(3+0)																		
	CS499	Computer Sc. Core (Final Year Project II)	3(3+0)																		
Total 133 Credit Hours																					

5.5. SEMESTER WISE COURSE DISTRIBUTION (BSCS – CYBER SECURITY)

Sem	Flag	Code	Course Title	CH	Pre-Req	Coverage															
Semester 1		CS111	Computer Sc. Core (Programming Fundamentals)	4(3+1)		<table><tr><th>Cat.</th><th>Courses</th><th>T. Credits</th></tr><tr><td> Comp. Core</td><td>1</td><td>4</td></tr><tr><td> GE</td><td>5</td><td>14</td></tr><tr><td>Total</td><td>6</td><td>18</td></tr></table>	Cat.	Courses	T. Credits	 Comp. Core	1	4	 GE	5	14	Total	6	18			
	Cat.	Courses	T. Credits																		
	 Comp. Core	1	4																		
	 GE	5	14																		
	Total	6	18																		
		ENG100	Gen. Edu. (Functional English)	3(3+0)																	
	CS101	Gen. Edu. (Applications of ICT)	3(2+1)																		
	PHY102	Gen. Edu. (NS - Applied Physics)	3(2+1)																		
	CS102	Gen. Edu. (QR1 - AMC I)	3(3+0)																		
	MAN101	Gen. Edu. (SS – Introduction to Management)	2(2+0)																		
Semester 2		CS112	Computer Sc. Core (Object Oriented Programming)	4(3+1)	CS111	<table><tr><th>Cat.</th><th>Courses</th><th>T. Credits</th></tr><tr><td> Comp. Core</td><td>2</td><td>8</td></tr><tr><td> GE</td><td>4</td><td>10</td></tr><tr><td>Total</td><td>6</td><td>18</td></tr></table>	Cat.	Courses	T. Credits	 Comp. Core	2	8	 GE	4	10	Total	6	18			
	Cat.	Courses	T. Credits																		
	 Comp. Core	2	8																		
	 GE	4	10																		
	Total	6	18																		
		CS113	Computer Sc. Core (Digital Logic Design)	4(3+1)																	
	ENG200	Gen. Edu. (Expository Writing)	3(3+0)																		
	CS103	Gen. Edu. (QR II - AMC II)	3(3+0)	CS102																	
	ISL100	Gen. Edu. (Islamic Studies / Ethics)	2(2+0)																		
	PKS100	Gen. Edu. (Pakistan Studies)	2(2+0)																		
Semester 3		CS211	Computer Sc. Core (Data Structures)	4(3+1)	CS111	<table><tr><th>Cat.</th><th>Courses</th><th>T. Credits</th></tr><tr><td> Comp. Core</td><td>2</td><td>8</td></tr><tr><td> IDS</td><td>1</td><td>3</td></tr><tr><td> GE</td><td>4</td><td>7</td></tr><tr><td>Total</td><td>7</td><td>18</td></tr></table>	Cat.	Courses	T. Credits	 Comp. Core	2	8	 IDS	1	3	 GE	4	7	Total	7	18
	Cat.	Courses	T. Credits																		
	 Comp. Core	2	8																		
	 IDS	1	3																		
	 GE	4	7																		
	Total	7	18																		
		CS212	Computer Sc. Core (Database Systems)	4(3+1)																	
	MATH101	IDS I (Calculus & Analytical Geometry)	3(3+0)																		
	PKS200	Gen. Edu. (Ideology and Constitution of Pakistan)	2(2+0)																		
	MGT200	Gen. Edu. (AH – Professional Practices)	2(2+0)																		
	SWS200	Gen. Edu. (Civics and Community Engagement)	2(2+0)																		
	ISL200	Gen. Edu. (Fehm-e-Quran - I)	1(0+1)																		
Semester 4		CS215	Computer Sc. Core (Operating Systems)	4(3+1)	CS101	<table><tr><th>Cat.</th><th>Courses</th><th>T. Credits</th></tr><tr><td> Comp. Core</td><td>4</td><td>13</td></tr><tr><td> GE</td><td>2</td><td>3</td></tr><tr><td>Total</td><td>2</td><td>16</td></tr></table>	Cat.	Courses	T. Credits	 Comp. Core	4	13	 GE	2	3	Total	2	16			
	Cat.	Courses	T. Credits																		
	 Comp. Core	4	13																		
	 GE	2	3																		
	Total	2	16																		
		CS213	Computer Sc. Core (Software Engineering)	3(3+0)																	
	CS216	Computer Sc. Core (Computer Networks)	3(3+0)																		
	CS214	Computer Sc. Core (Design & Analysis of Algorithm)	3(3+0)																		
	MGT400	Gen. Edu. (Entrepreneurship)	2(2+0)																		
	ISL300	Gen. Edu. (Fehm-e-Quran II)	1(0+1)																		
Semester 5		CS311	Computer Sc. Core (Artificial Intelligence)	3(3+0)	CS101	<table><tr><th>Cat.</th><th>Courses</th><th>T. Credits</th></tr><tr><td> Comp. Core</td><td>2</td><td>6</td></tr><tr><td> Sp. Elective</td><td>2</td><td>6</td></tr><tr><td> IDS</td><td>2</td><td>6</td></tr><tr><td>Total</td><td>6</td><td>18</td></tr></table>	Cat.	Courses	T. Credits	 Comp. Core	2	6	 Sp. Elective	2	6	 IDS	2	6	Total	6	18
	Cat.	Courses	T. Credits																		
	 Comp. Core	2	6																		
	 Sp. Elective	2	6																		
	 IDS	2	6																		
	Total	6	18																		
	CS312	Computer Sc. Core (Theory of Automata)	3(3+0)	CS103																	
	CS351	Sp. Elec. I (Wireless and Mobile Security)	3(3+0)																		
	CS342	Sp. Elec. II (Machine Learning)	3(3+0)																		
	MATH111	IDS II (Linear Algebra)	3(3+0)																		
	MATH198	IDS III (Discrete Structure)	3(3+0)																		
Semester 6		CS411	Computer Sc. Core (Information Security)	3(3+0)	CS103	<table><tr><th>Cat.</th><th>Courses</th><th>T. Credits</th></tr><tr><td> Comp. Core</td><td>2</td><td>6</td></tr><tr><td> Sp. Elective</td><td>2</td><td>6</td></tr><tr><td> IDS</td><td>1</td><td>3</td></tr><tr><td>Total</td><td>5</td><td>15</td></tr></table>	Cat.	Courses	T. Credits	 Comp. Core	2	6	 Sp. Elective	2	6	 IDS	1	3	Total	5	15
	Cat.	Courses	T. Credits																		
	 Comp. Core	2	6																		
	 Sp. Elective	2	6																		
	 IDS	1	3																		
	Total	5	15																		
	CS412	Computer Sc. Core (Computer Org & Architecture)	3(2+1)	CS113																	
	CS353	Sp. Elec. III (Secure Software Design and Dev.)	3(3+0)																		
	CS354	Sp. Elec. IV (Digital Forensics and Incident Response)	3(3+0)																		
	STAT300	IDS IV (Probability & Statistics)	3(3+0)																		
Semester 7		CS413	Computer Sc. Core (Cloud Computing)	3(3+0)	CS101	<table><tr><th>Cat.</th><th>Courses</th><th>T. Credits</th></tr><tr><td> Comp. Core</td><td>1</td><td>3</td></tr><tr><td> Sp. Elective</td><td>2</td><td>6</td></tr><tr><td> Professional</td><td>2</td><td>6</td></tr><tr><td>Total</td><td>5</td><td>15</td></tr></table>	Cat.	Courses	T. Credits	 Comp. Core	1	3	 Sp. Elective	2	6	 Professional	2	6	Total	5	15
	Cat.	Courses	T. Credits																		
	 Comp. Core	1	3																		
	 Sp. Elective	2	6																		
	 Professional	2	6																		
	Total	5	15																		
	CS451	Sp. Elec. V (Network Security)	3(3+0)																		
	CS452	Sp. Elec. VI (Applied Cryptography)	3(3+0)																		
	CS497	Field Experience (Internship)	3(3+0)																		
	CS498	Computer Sc. Core (Final Year Project I)	3(3+0)																		
Semester 8		CS453	Sp. Elec. VII (Usable Security and Privacy)	3(3+0)		<table><tr><th>Cat.</th><th>Courses</th><th>T. Credits</th></tr><tr><td> Sp. Elective</td><td>3</td><td>9</td></tr><tr><td> Professional</td><td>2</td><td>6</td></tr><tr><td>Total</td><td>5</td><td>15</td></tr></table>	Cat.	Courses	T. Credits	 Sp. Elective	3	9	 Professional	2	6	Total	5	15			
	Cat.	Courses	T. Credits																		
	 Sp. Elective	3	9																		
	 Professional	2	6																		
	Total	5	15																		
	CS454	Sp. Elec. VIII (Cloud Security)	3(3+0)																		
	CS455	Sp. Elec. IX (Artificial Intelligence for Cyber Security)	3(3+0)																		
	CS496	Professional Certification	3(3+0)																		
	CS499	Computer Sc. Core (Final Year Project II)	3(3+0)																		
Total 133 Credit Hours																					

5.6. SEMESTER WISE COURSE DISTRIBUTION (ASSOCIATE DEGREE)

Sem	Flag	Code	Course Title	CH	Pre-Req	Coverage																	
Semester 1		CS111	Comp. Sc. Core (Programming Fundamentals)	4(3+1)		<table><tr><th>Cat.</th><th>Courses</th><th>T. Credits</th></tr><tr><td> Comp. Core</td><td>1</td><td>4</td></tr><tr><td> GE</td><td>5</td><td>14</td></tr><tr><td>Total</td><td>6</td><td>18</td></tr></table>			Cat.	Courses	T. Credits	 Comp. Core	1	4	 GE	5	14	Total	6	18			
	Cat.	Courses	T. Credits																				
	 Comp. Core	1	4																				
	 GE	5	14																				
	Total	6	18																				
		ENG100	Gen. Edu. (Functional English)	3(3+0)																			
	CS101	Gen. Edu. (Applications of ICT)	3(2+1)																				
	PHY102	Gen. Edu. (NS - Applied Physics)	3(2+1)																				
	CS102	Gen. Edu. (QR1 - AMC I)	3(3+0)																				
	MAN101	Gen. Edu. (SS – Introduction to Management)	2(2+0)																				
Semester 2		CS112	Comp. Sc. Core (Object Oriented Programming)	4(3+1)	CS111	<table><tr><th>Cat.</th><th>Courses</th><th>T. Credits</th></tr><tr><td> Comp. Core</td><td>1</td><td>4</td></tr><tr><td> Elective</td><td>1</td><td>4</td></tr><tr><td> GE</td><td>4</td><td>10</td></tr><tr><td>Total</td><td>6</td><td>18</td></tr></table>			Cat.	Courses	T. Credits	 Comp. Core	1	4	 Elective	1	4	 GE	4	10	Total	6	18
	Cat.	Courses	T. Credits																				
	 Comp. Core	1	4																				
	 Elective	1	4																				
	 GE	4	10																				
	Total	6	18																				
	CS113	Elective I (Digital Logic Design)	4(3+1)																				
	ENG200	Gen. Edu. (Expository Writing)	3(3+0)																				
	CS103	Gen. Edu. (QR II - AMC II)	3(3+0)	CS102																			
	ISL100	Gen. Edu. (Islamic Studies / Ethics)	2(2+0)																				
	PKS100	Gen. Edu. (Pakistan Studies)	2(2+0)																				
Semester 3		CS211	Computer Sc. Core (Data Structures)	4(3+1)	CS111	<table><tr><th>Cat.</th><th>Courses</th><th>T. Credits</th></tr><tr><td> Comp. Core</td><td>2</td><td>8</td></tr><tr><td> Elective</td><td>1</td><td>3</td></tr><tr><td> GE</td><td>4</td><td>7</td></tr><tr><td>Total</td><td>7</td><td>18</td></tr></table>			Cat.	Courses	T. Credits	 Comp. Core	2	8	 Elective	1	3	 GE	4	7	Total	7	18
	Cat.	Courses	T. Credits																				
	 Comp. Core	2	8																				
	 Elective	1	3																				
	 GE	4	7																				
	Total	7	18																				
		CS212	Computer Sc. Core (Database Systems)	4(3+1)																			
		CS261	Elective II (Web Development) 	3(3+0)																			
	PKS200	Gen. Edu. (Ideology and Constitution of Pakistan)	2(2+0)																				
	MGT200	Gen. Edu. (AH – Professional Practices)	2(2+0)																				
	SWS200	Gen. Edu. (Civics and Community Engagement)	2(2+0)																				
	ISL200	Gen. Edu. (Fehm-e-Quran - I)	1(0+1)																				
Semester 4		CS215	Computer Sc. Core (Operating Systems)	4(3+1)	CS101	<table><tr><th>Cat.</th><th>Courses</th><th>T. Credits</th></tr><tr><td> Comp. Core</td><td>3</td><td>10</td></tr><tr><td> GE</td><td>2</td><td>3</td></tr><tr><td> Elective</td><td>2</td><td>6</td></tr><tr><td>Total</td><td>7</td><td>19*</td></tr></table>			Cat.	Courses	T. Credits	 Comp. Core	3	10	 GE	2	3	 Elective	2	6	Total	7	19*
	Cat.	Courses	T. Credits																				
	 Comp. Core	3	10																				
	 GE	2	3																				
	 Elective	2	6																				
	Total	7	19*																				
		CS213	Computer Sc. Core (Software Engineering)	3(3+0)																			
	CS216	Computer Sc. Core (Computer Networks)	3(3+0)																				
	CS214	Elective III (Design & Analysis of Algorithms)	3(3+0)																				
	CS262	Elective IV (Intro to ML and AI Tools) 	3(3+0)																				
	ISL300	Gen. Edu. (Entrepreneurship)	2(2+0)																				
	CS111	Gen. Edu. (Fehm-e-Quran II)	1(0+1)																				

Total 73 Credit Hours

*Courses are aligned with the regular BS program. Only two courses are different and are indicated by . Students opting for an Associate Degree may take these two courses in the 3rd and 4th semesters, or both in the 4th or summer semesters, enabling exit with an Associate Degree Computing.

**Semester 4 has 19 credit hours, and the Board of Studies, with the approval of the Dean of Faculty, has authorized the Chairman to offer this semester for students opting for an Associate Degree in Computing with 19 CH.

Prepared By:

Dr. M. Irfan Uddin

Associate Professor

Department of Computer Science, AWKUM.

Approved By:

Prof. Dr. Nadeem Iqbal

Chairman

Department of Computer Science, AWKUM.